



0013

## TEST REPORT

**Lucideon Reference:** N26458 (QT-78966/4/KNA)/Ref. 3

**Project Title:** Shear Testing to BS EN 1052-3:2002 of N10 Carbon Negative Bricks

**Client:** Earth4Earth Technology Ltd  
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**For the Attention of:** Ms Anna Simmons

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**Report Date:** 26 May, 2026

**Purchase Order No.:** PO-0003

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Opinions and interpretations expressed herein are outside the scope of UKAS Accreditation.

Mr Justin Fryer  
**Testing Team  
Reviewer**

Mr Kaleem Zar  
**Testing Team  
Project Manager**



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## 1 INTRODUCTION

Earth4Earth Technology Ltd required a programme of testing to determine the shear strength of masonry samples comprising of N10 Carbon Negative bricks in accordance with BS EN 1052-3:2002 Methods of Test for Masonry – Part 3: Determination of Initial Shear Strength.

Testing was carried out on 7 May 2026 at Lucideon's Structures Laboratory at Queens Road, Penkhull, Stoke-on-Trent, ST4 7LQ.

## 2 SAMPLES RECEIVED

Solid bricks referenced as N10 Carbon Negative were received from the client.

The bricks were nominally 215 mm x 102.5 mm x 65 mm (length x width x height).

The bricks were solid with a nominal compressive strength of 39.1 N/mm<sup>2</sup>.

## 3 METHOD OF CONSTRUCTION

Nine triplet samples were constructed by a Lucideon Limited with M4 mortar applied at each course.

The Mortar had a compressive strength of 4.16 N/mm<sup>2</sup>.

Following construction, the samples were pre-compressed by a uniform mass to give a vertical stress of  $2.07 \times 10^{-3}$  N/mm<sup>2</sup>.

The samples were close covered with polythene sheets for 28 days at ambient laboratory temperatures.

The samples were constructed on 9 April 2026.

## 4 METHOD OF TEST

The sample was placed in the shear test rig, such that, the faces of the masonry units, where the shear load is applied, were plane and perpendicular to the direction of the shear load.

The rate of applied load was in between 0.1 N/mm<sup>2</sup> per minute and 0.4 N/mm<sup>2</sup> per minute, which was within the limits given in BS EN 1052-3:2002.

Three tests at each of the precompressions of 0.2, 0.6 and 1.0 N/mm<sup>2</sup> were carried out following Procedure A in BS EN 1052-3:2002.

The testing was carried out under ambient laboratory temperature and humidity.

## 5 RESULTS

Individual shear results are shown in the Tables Section.

A plot of shear strength versus precompression can be seen for every test in the Chart Section.

The failure mode is shown in Plate 1.

## **6 SUMMARY**

The summary of results calculated based on data recorded during testing can be seen in the Tables Section.

## TABLES

Table 1 – Mortar Properties

| *Compressive Strength<br>to EN 1015-11<br>(N/mm <sup>2</sup> ) * | *Air Content to EN 1015-7<br>(%) | *Flow Value to BS 4551 |
|------------------------------------------------------------------|----------------------------------|------------------------|
| 4.16                                                             | 5.1                              | 174                    |

*\*These tests are not on Lucideon's UKAS Schedule.*

Table 2 – Shear Results for N10 Carbon Negative Bricks Tested in Accordance with BS EN 1052-3:2002

| Sample No. | $F_{i,max}$ Maximum<br>Shear Load<br>(N) | Length<br>(mm) | Width<br>(mm) | $f_{pi}$ Precompressive Stress of<br>Individual Sample<br>(N/mm <sup>2</sup> ) | $f_{voi}$ Shear Strength of<br>Individual Sample<br>(N/mm <sup>2</sup> ) | Failure Mode                          |
|------------|------------------------------------------|----------------|---------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------|
| 1          | 8702                                     | 215            | 100           | 0.2                                                                            | 0.20                                                                     | A1 in accordance with<br>BS EN 1052-3 |
| 2          | 9409                                     | 215            | 100           |                                                                                | 0.22                                                                     |                                       |
| 3          | 9002                                     | 215            | 102           |                                                                                | 0.21                                                                     |                                       |
| 4          | 21010                                    | 216            | 103           | 0.6                                                                            | 0.47                                                                     |                                       |
| 5          | 19540                                    | 216            | 103           |                                                                                | 0.44                                                                     |                                       |
| 6          | 20360                                    | 215            | 102           |                                                                                | 0.46                                                                     |                                       |
| 7          | 28950                                    | 215            | 103           | 1.0                                                                            | 0.65                                                                     |                                       |
| 8          | 29340                                    | 216            | 105           |                                                                                | 0.65                                                                     |                                       |
| 9          | 27540                                    | 215            | 105           |                                                                                | 0.61                                                                     |                                       |

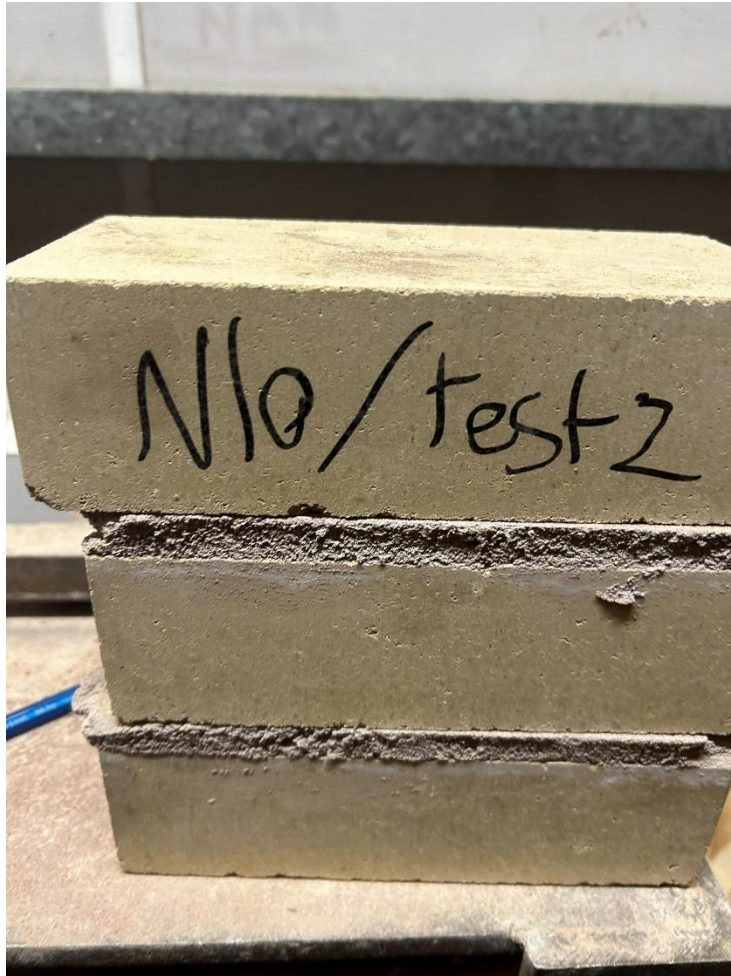
**Table 3** – The Summary of Testing Results for N10 Carbon Negative Bricks Tested in Accordance with BS EN 1052-3:2002

| Name of the Characteristic                             | Value                  |
|--------------------------------------------------------|------------------------|
| Mean initial shear strength was determined             | 0.11 N/mm <sup>2</sup> |
| Internal angle of friction                             | 28.15°                 |
| Characteristic initial shear strength<br>(Procedure A) | 0.09 N/mm <sup>2</sup> |
| Characteristic angle of internal friction              | 23.17°                 |

**NOTE:** The results given in this report apply only to the samples that have been tested.

**END OF REPORT**

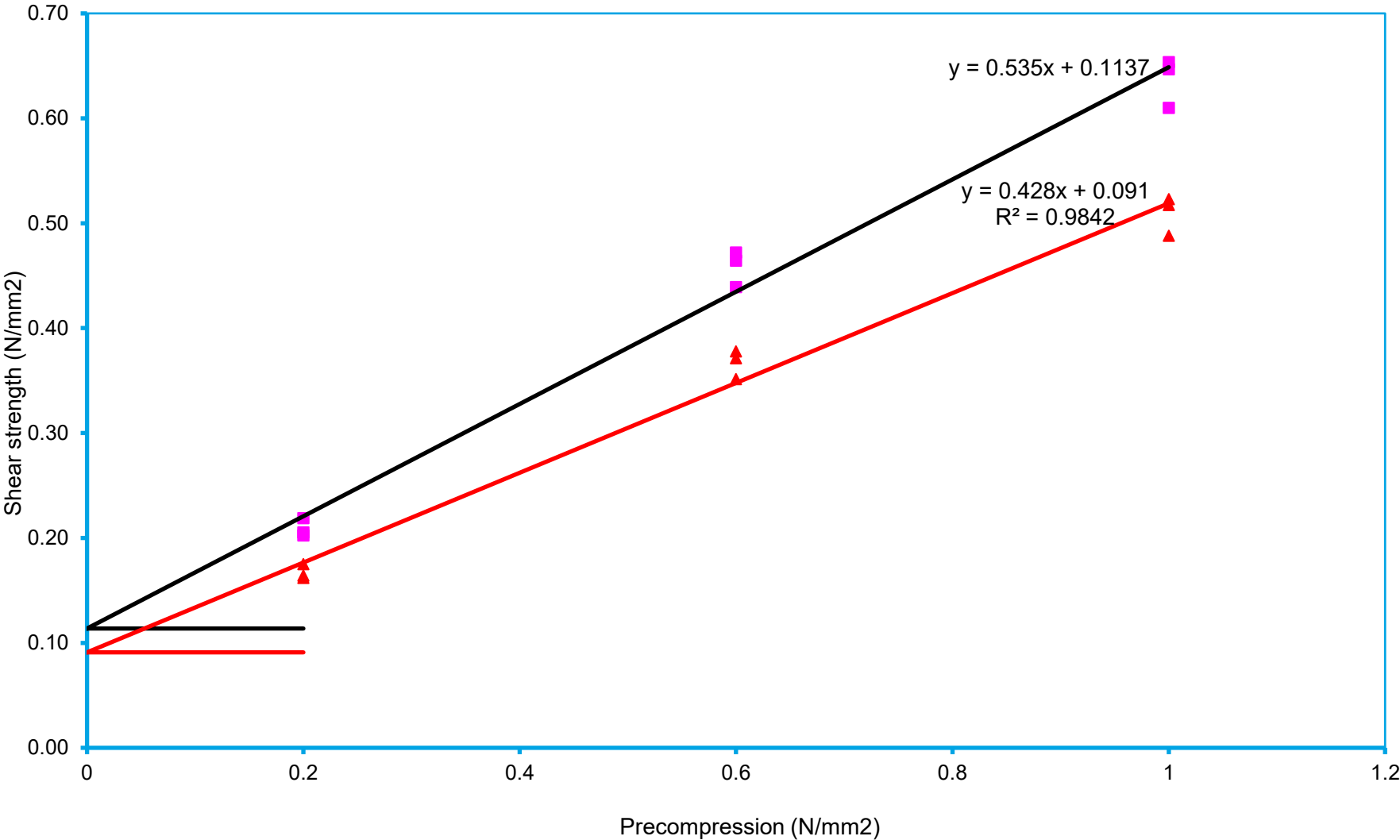
**PLATE**



**Plate 1** – Typical Shear Test Failure

**Chart 1 - Shear Strength vs Pre-Compression Chart for N10 Carbon Negative Bricks**

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Key

- Linear ( )
- Shear Strength
- Characteristic shear strength
- Linear ( )